

SOLUTION

Question 1: Solution – A Search Algorithm*

The A* Search Algorithm starts from node A and uses the evaluation function $f(n) = g(n) + h(n)$, where $g(n)$ is the actual path cost from the start node and $h(n)$ is the heuristic estimate to the goal. Initially, node A is placed in the Open List with $g(A)=0$, $h(A)=7$, and $f(A)=7$. After expanding A, the neighboring nodes B and C are added to the Open List with $f(B)=8$ and $f(C)=8$.

Next, node B is selected for expansion because it has the lowest evaluation value. Expanding B generates nodes D and E. Node E is reached with $g(E)=4$ and $f(E)=6$, making it the next node to be expanded. From E, a shorter path to D is found, reducing its cost to $g(D)=6$ and $f(D)=9$. Node C is then expanded, but it does not provide a better path to any node. Afterward, node D is expanded, which leads directly to the goal node G with $g(G)=8$ and $f(G)=8$. Since G is the goal node, the search terminates.

The optimal path obtained is $A \rightarrow B \rightarrow E \rightarrow D \rightarrow G$, and the total path cost is 8. The A* algorithm successfully identifies the least-cost path by combining the actual path cost with heuristic information, resulting in an efficient search process.

Question 2: Solution – Minimax with Alpha–Beta Pruning

The Minimax algorithm begins at the MAX node with initial values $\alpha = -\infty$ and $\beta = +\infty$. The left MIN subtree is evaluated first. Its leaf nodes have values 3, 5, and 6, so the MIN node selects the smallest value, which is 3. This value is returned to the MAX node, updating $\alpha = 3$.

The algorithm then evaluates the right MIN subtree. The first leaf node has value 9, so $\beta = 9$. The second leaf node has value 1, updating $\beta = 1$. At this point, the condition $\alpha \geq \beta$ ($3 \geq 1$) becomes true, so the

remaining leaf node with value 2 is pruned because it cannot affect the final decision.

Finally, the MAX node compares the values returned from both MIN nodes, which are 3 and 1, and selects the larger value 3 as the final minimax value. Therefore, the best move for the MAX player is the left subtree, the final minimax value is 3, and the leaf node with value 2 is pruned using Alpha–Beta Pruning.